

Zeen Zheng

Portfolio Website: zzez1127.github.io/my_portfolio_website/

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EDUCATION

University of Washington

Seattle, WA

Bachelor of Science in Computer Science

Expected Jun 2027

- **Relevant Coursework:** Linear Algebra, Computer Programming II/III (accelerated), System And Software Tools, Scientific Computing, Hardware/Software Interface, Foundations of Computing I & II, Data Structure and Parallelism, Design Methods

TECHNICAL SKILLS

Languages/Skills: Python, Java, SQL, C, C#, JavaScript, Dart, HTML/CSS, Swift, $\text{\LaTeX}$

Frameworks: PySpark, FastAPI, React, Node.js, Scikit-learn, XGBoost, Unity Engine, Flutter

Tools: Databricks, Git, Docker, Linux, Bash, VS Code, Figma, AWS (Bedrock, Lambda)

Libraries: pandas, NumPy, SHAP, Matplotlib, tkinter, Swing

EXPERIENCE

NRG Information Technology Team

Sept 2023 – Jun 2025

Newport Robotics Group (NRG 948)

Bellevue, WA

- Enhanced competition data collection efficiency by **40%** through the development and deployment of a **cross-platform scouting app** using **Flutter** for a 50-member team.
- Reduced data entry errors by **15%** by iteratively optimizing the UI/UX based on real-time feedback from scouts during competition heats.

Member of SGAI at Beaver Works Summer Institute

Jul 2024 – Aug 2024

Massachusetts Institute of Technology

Cambridge, MA

- Mastered systems modeling and AI ethics in an intensive **8-week** program; applied concepts to game mechanics design.
- Collaborated in an Agile team of five to implement game components using **Python**, **Unity**, and **Git**.
- Led AI algorithm implementation and performance optimization for a team-based game project.

UI Designer & AI Lead (Team 24)

Oct 2024

UW WINFO Hackathon

Seattle, WA

- Won the **"Best Use of AWS AI"** and **Honorable Mention** among 30+ teams.
- Built UI prototypes and marketing visuals for an AI-powered carbon footprint tracker using **Figma** and **AWS cloud services** to promote environmental awareness.

PROJECTS

Weather-Yield Detection Model (Databricks Hackathon) | *PySpark, Databricks, FastAPI*

Mar 2026

- Engineered a distributed data pipeline using **PySpark** on **Databricks** to process and merge millions of rows of NOAA daily weather data and RMA crop yield reports.
- Trained an **XGBoost** regression model with Grouped K-Fold cross-validation to predict baseline crop yields, using **SHAP** values to extract feature-level explanations for yield anomalies (Z-score > 2.0).
- Deployed the model via a **FastAPI** backend and integrated the **OpenAI Assistants API** to allow stakeholders to query complex data insights using natural language.

Autonomous Agent Behavior Prediction | *Python, Unity, CNNs, Machine Learning*

Jul 2024 – Aug 2024

- Attained a **92% classification accuracy** across 5 CNN models by generating a custom dataset of **2,000+ synthetic images** in Unity to train neural networks.
- Enhanced object recognition speed by **18%** by optimizing Unity-to-Python data pipelines and refining model hyperparameters for real-time inference.

Classic Games Hub | *Java, HTML/CSS, UI, Full-stack*

May 2025 – Jun 2025

- Achieved **100% logic accuracy** in a Java-based Minesweeper clone by developing recursive tile-clearing algorithms and a custom event-handling system.
- Boosted user retention by **15%** (simulated) by designing a responsive web interface and integrating smooth UI transitions between different game modules.

CERTIFICATIONS

Cisco Certified Network Associate (CCNA 200-301), Cisco

Jun 2025